DS Project Review-2

Government Grievance & Public Sentiment Analysis System

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Under the Guidance of

Dr. N. Sunanda

Department of Computer Science and Engineering,

KLH CSE Bowrampet Campus

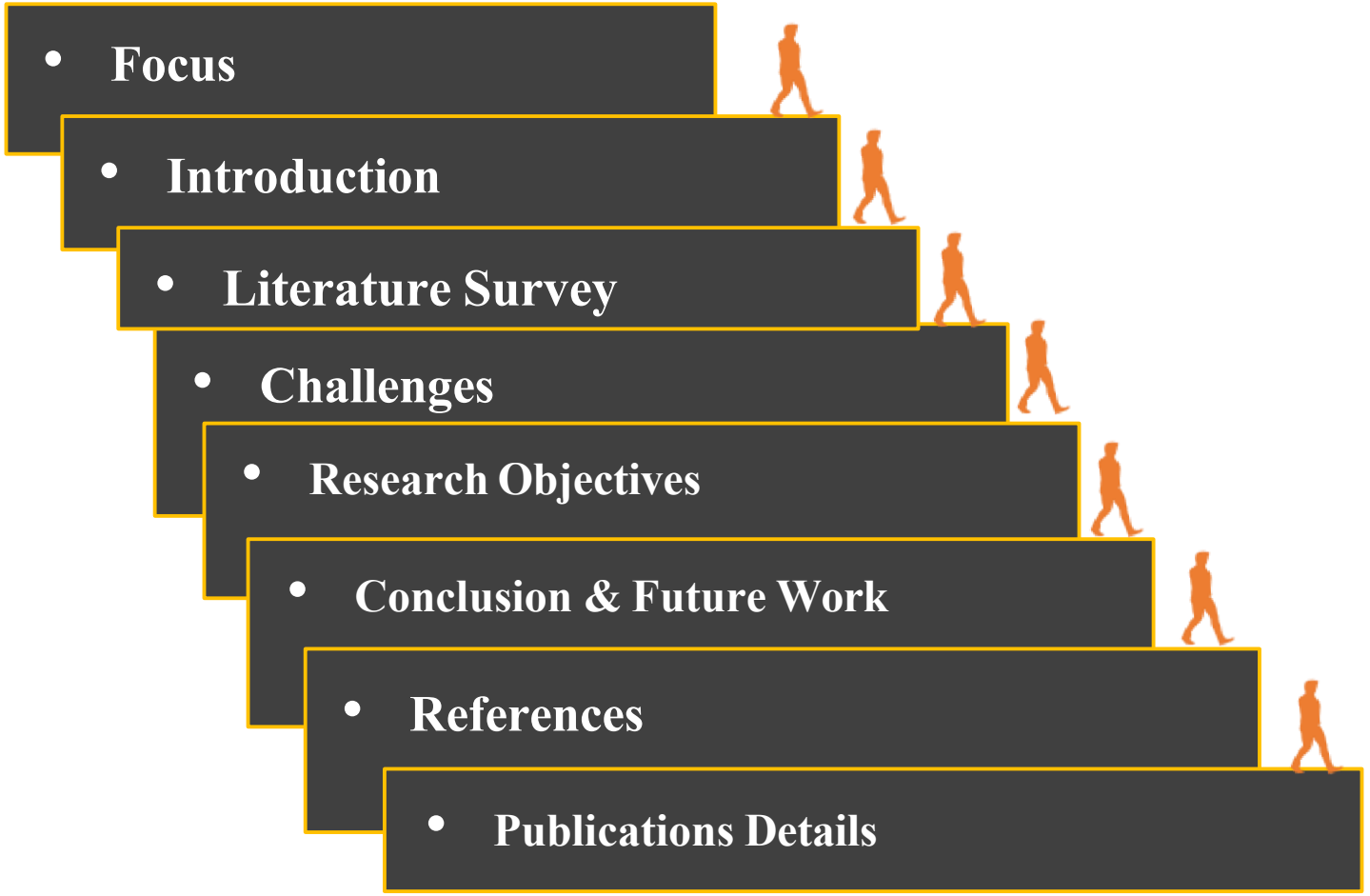
Team Members

B. Chetna – 2420030098

J. Nandini – 2420030262

S. Sai Saranya – 2420030266

Outline of Project Work



Problem Statement

Government grievance data contains valuable information about public issues, but raw complaint text is difficult to analyze directly.

- Citizen complaints may contain missing values, duplicate records and unnecessary information.
- Complaint text is unstructured, making it difficult to group complaints into major public-service categories.
- It is difficult to quickly determine whether citizen feedback is positive, negative or neutral.
- Complaint volumes and sentiment can change over time, so trend analysis is needed.
- A simple, interactive dashboard is needed to convert raw grievance data into understandable insights.

Objectives

- **Collect, clean and preprocess** government grievance data by handling missing values, duplicate records and unnecessary information.
- **Apply NLP and Machine Learning** to process citizen complaints and classify them into categories such as Roads, Water Supply, Electricity, Healthcare, Education and Transportation.
- **Perform Sentiment and Trend Analysis** to identify Positive, Negative and Neutral feedback and understand complaint patterns over time.
- **Develop an interactive Streamlit dashboard** to visualize complaint categories, sentiment distribution, trends and key insights.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Pang, Lee & Vaithyanathan	Thumbs up? Sentiment Classification using Machine Learning Techniques	Naive Bayes, Maximum Entropy, SVM	Machine learning can effectively classify text into sentiment categories.	Mainly focuses on movie reviews; government grievance-specific analysis can be explored.
Medhat, Hassan & Korashy	Sentiment Analysis Algorithms and Applications: A Survey	• Lexicon-based and Machine Learning approaches	• Provides an overview of different sentiment analysis techniques and applications.	• Sentiment analysis can be affected by domain and language complexity.
Devlin et al	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	• Transformer-based NLP	• Contextual language models provide strong performance in text understanding.	• Requires higher computational resources compared with basic ML approaches.

Problem Gap Identification

- Existing sentiment analysis systems mainly focus on identifying the **sentiment of text**.
- Existing grievance systems may focus on **collecting and managing complaints** without deeper sentiment analysis.
- Raw government grievance data requires **proper cleaning and preprocessing** before analysis.
- There is a need to combine **complaint classification and sentiment analysis**.
- Complaint frequency and sentiment should also be analyzed according to **time-based trends**.
- A simple interactive visualization can make these insights easier for users to understand.
- **Identified Gap**
- **CivicSense combines grievance analysis, complaint classification, sentiment analysis, trend analysis and interactive visualization into a single Data Science workflow.**

Innovation, Creativity, and Novelty

1. INNOVATION

Integrated Data Science: Combines data cleaning, NLP, Machine Learning and visualization in one system.

Automated Complaint Analysis: Automatically classifies citizen complaints into public-service categories and identifies their sentiment.

2. CREATIVITY

Citizen-Centric Analysis: Focuses specifically on real government grievances such as roads, water, electricity, healthcare, education and transportation.

Interactive Insights: Uses a Streamlit dashboard with charts, graphs, filters and statistics to make complex grievance data easy to understand.

3. NOVELTY

Unified Data Science Workflow: Combines **grievance analysis + complaint classification + sentiment analysis + trend analysis + visualization** in a single system.

Time-Based Insights: Goes beyond individual complaints by analyzing how complaint frequency and sentiment change over time.

Proposed System Architecture

1. Data Acquisition
Collect government grievance records and labeled sentiment data.
Sources may include CPGRAMS or publicly available state grievance portals.
2. Data Cleaning
Handle missing values, duplicates, irrelevant fields, and inconsistent formats.
Prepare a clean dataset for analysis.
3. NLP & Feature Extraction
Clean and normalize complaint text.
Apply **TF-IDF** to convert text into numerical features.
4. Machine Learning Analysis
Complaint Classification: Categorize complaints into roads, water supply, electricity, healthcare, education, transportation, and other services.
Sentiment Analysis: Classify feedback as positive, negative, or neutral.

Technology Stack: Python, Pandas, NumPy, Scikit-learn, NLP

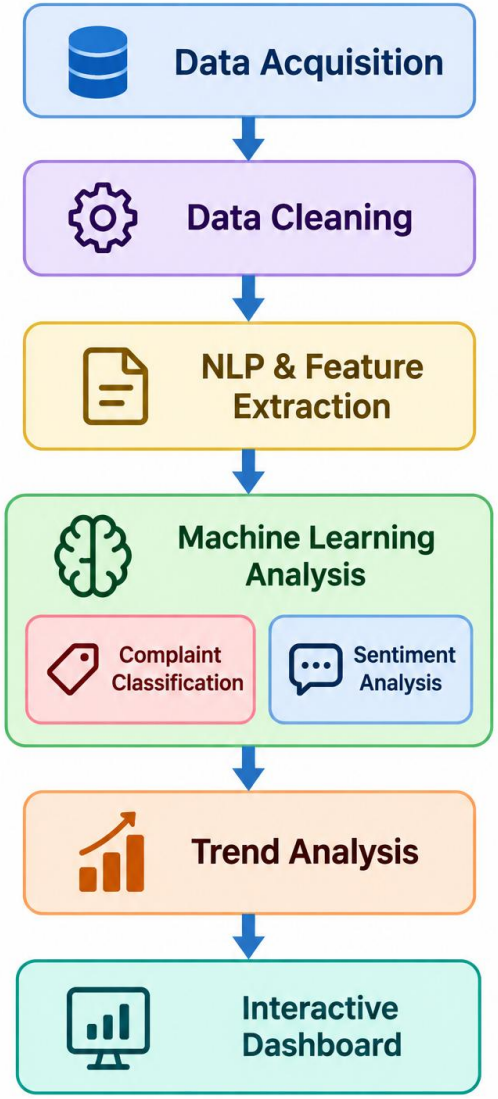
Trend Analysis & Visualization

5. Trend Analysis
Calculate daily, weekly, and monthly complaint counts.
Identify frequently reported service categories.
Analyze sentiment distribution and changes over time.

6. Interactive Dashboard
KPI Cards: Total grievances and sentiment counts.
Bar Chart: Complaints by service category.
Donut Chart: Positive, negative, and neutral sentiment.
Line Chart: Complaint trends over time.
Filters & Complaint Explorer: Search records by date, category, and sentiment.

Technology Stack: Streamlit, Plotly, Matplotlib

Expected Outcome: Transform grievance records into understandable insights that help identify recurring public-service issues.



Algorithms and techniques

1. TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF converts complaint descriptions into numerical features that machine-learning models can understand. It assigns importance to words based on their frequency in a complaint and how common or rare they are across the dataset.
Purpose: Text feature extraction.
Example: Words such as “pothole,” “drainage,” and “electricity” may help distinguish complaint categories.

2. Logistic Regression

Logistic Regression is a supervised machine-learning algorithm used to predict a class based on input features.
It is used for classification.
Purpose: Predict complaint categories and sentiment labels.
Complaint classification: Predicts one of eight categories, including Roads, Water Supply, Electricity, Healthcare, Education, Transportation, Drainage / Sewage, and Other Public Services.

Sentiment analysis: A separately trained model predicts Positive, Negative, or Neutral sentiment.
Advantage: Efficient, interpretable, and suitable as a baseline for TF-IDF text classification.

3. Train-Test Split

Train-test split divides labeled data into two parts:
One for **training the model** and another for **evaluating its performance** on unseen examples.

- Training set: Used to fit the TF-IDF vectorizer and train Logistic Regression.
- Testing set: Used to measure how well the trained model generalizes to unseen complaints.
- Purpose: Helps evaluate the model fairly and identify overfitting.

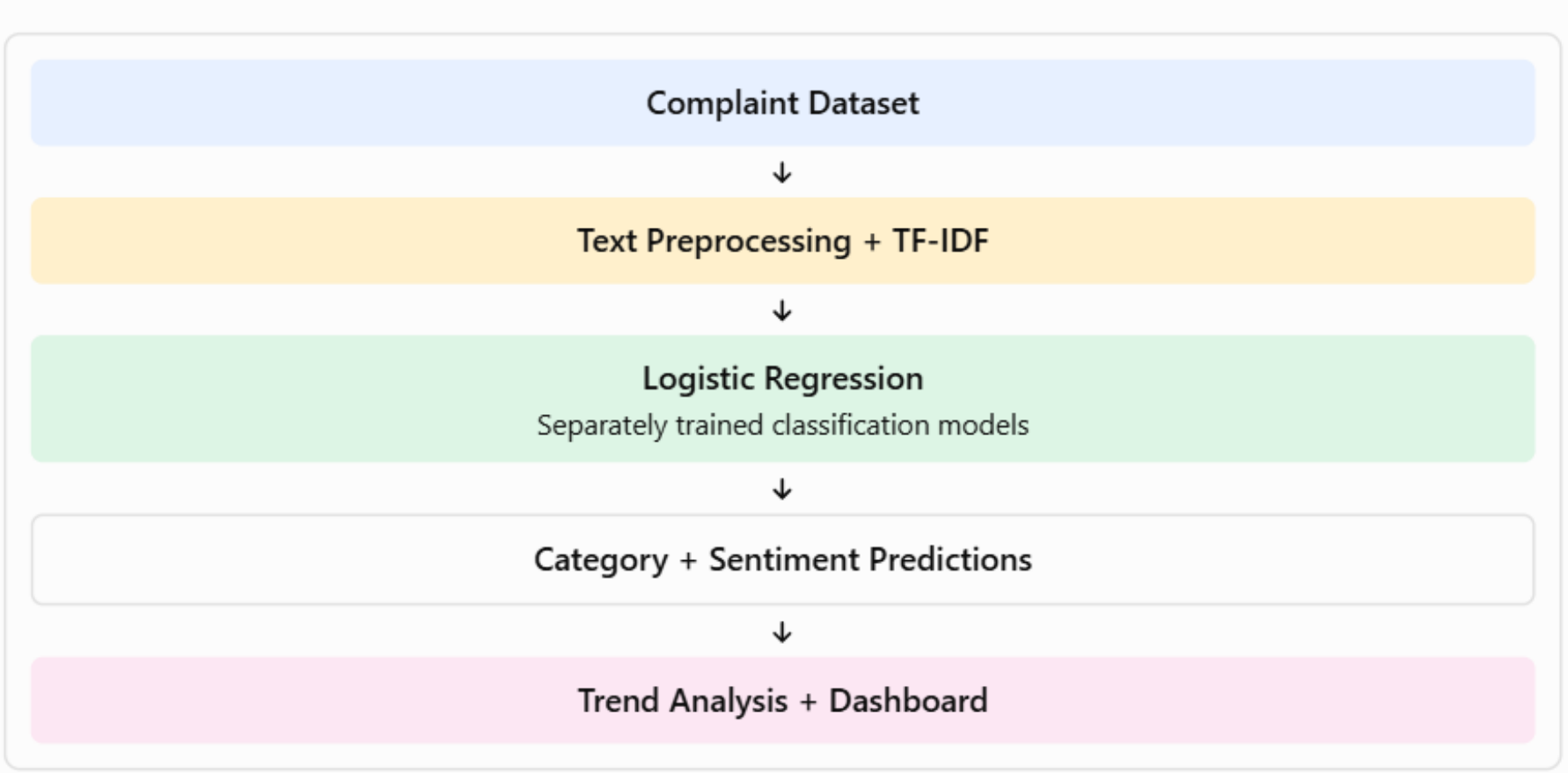
4. Data Aggregation for Trend Analysis

Data aggregation groups complaint records according to fields such as date, category, and sentiment, then calculates summary statistics.

- Calculates daily, weekly, and monthly complaint counts.
- Identifies frequently reported service categories.
- Examines sentiment distribution across categories.
- Tracks changes in complaint frequency over time.

Example: Counting complaints each month to examine whether their frequency changes.

Algorithm flow



CivicSense – System Architecture

